AsOf: Auditable Source Reporting for Therapeutic Evidence Workflows

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Abstract

Biomedical agent reports can preserve a citation while changing a number, importing later information or concealing an unavailable field. We present AsOf, a versioned evidence-workflow implementation, and a bounded technical study of literal source reporting. The study tests source fidelity and admission policy, rather than therapeutic judgment. Before execution, we fixed twelve units selected by identity from archived Open Targets 25.09 data, source-only generation inputs, separately reconstructed references, an all-output schedule and a descriptive analysis. Each unit requests eleven identity fields and four native measurement fields; forty of forty-eight measurement slots are populated. A deterministic literal assembler produces verified exact reports in twelve of twelve units and completes every answerable measurement group at the frozen 60%, 80% and 100% thresholds. An always-abstain control completes zero of twelve. In fourteen exposed constructed scenarios, the full policy admits none of ten deliberately invalid reports and rejects none of four technically valid reports. Removing the study-specific eligibility wrapper admits two invalid scenarios; retaining only byte anchoring and output-schema checks admits seven. All thirty-eight outputs and ninety policy decisions are retained. Production takes 203.828 seconds, followed by a separate 7.125-second source-inclusive audit. Earlier local generator probes failed their declared contracts and remain negative development evidence. These findings demonstrate a reconstructable technical reporting path on this selected set. They do not establish clinical accuracy, causal genetic interpretation, independent-study generalization, model superiority or provider authenticity. Numerical prediction features remain empty and clinical probabilities null.

1. Introduction

Therapeutic assessment combines sources with different release dates, identifiers, biological contexts and permissions. A report can be faithful to its retained bytes yet use an ineligible source or convey an unsupported conclusion. A missing measurement is also different from an absent biological effect. An auditable workflow therefore needs to distinguish byte integrity, mechanical consistency and scientific validity.

Biomedical agents already access domain tools and coordinate complex tasks. GeneGPT uses genomic tools [1]; Biomni reports broad biomedical workflow execution [2]; The Virtual Biotech organizes agents around therapeutic-development functions [3]. Existing approaches also evaluate evidence support. Alvessa checks generated statements against retrieved records and exercises corruption controls [4]. ALCE separates answer and citation quality [5], ARES evaluates retrieval-augmented systems with learned judges and human calibration [6], and biomedical quotation assessment distinguishes quotations from supported interpretation [7]. Structured deterministic grading and process-level evaluation are represented by TxBench-PP [8] and BiomniBench [9]. AsOf does not establish priority for any of these individual ideas.

Our narrower question is whether a local implementation can preserve a declared source-reporting contract through source selection, normalization, generation, reference construction, scoring and retained execution records. The implemented contract treats source availability, hypothesis scope, evaluator ancestry and exact field bindings as first-class constraints. We evaluate native-field copying and deliberately designed failures; we do not evaluate target ranking, experiment design, mechanism inference or therapeutic recommendations.

The contribution is an inspectable combination of versioned evidence admission, immutable source-bound records, explicit missingness, separate generation and evaluation roles, durable execution accounting and source-inclusive replay. Its empirical support is a small technical feasibility study and a negative model-development result. The perfect literal baseline is an expected ceiling for a copying task. It is evidence that this reporting path works on the selected inputs, not evidence of biomedical reasoning or a general advantage over other agents.

2. Implementation and threat model

2.1 Evidence, time and hypothesis scope

AsOf separates cases, source snapshots, evidence packets, input-lineage nodes and execution artifacts. The default Python runtime has no third-party dependencies; actual-format Parquet ingestion uses the optional pinned PyArrow 21.0.0 dependency. Version-two evidence is copied and recursively frozen on admission. Ledger commitments are checked against admitted canonical fingerprints. Features bind to a hypothesis fingerprint covering canonical target and indication identifiers, namespace versions, intended action, modality and tissue. Raw observations can remain contextual when a numerical feature is inapplicable.

Information availability is separate from computation time. A result newly computed from eligible older inputs can qualify; an ancestor containing later information or an evaluation-only artifact cannot. Mapping and annotation parents belong in the same eligibility graph. Public download dates provide present-day upper bounds on observed availability, not proof of historical first availability. Null dates or unresolved source context are not silently supplied from a newer release. The present experiment uses a contemporary cutoff and does not establish a historically uncontaminated clinical forecast.

The implementation checks declared identifiers, roles, dates, hashes and source contents under trusted local configuration. It does not authenticate a provider's original assertion, defend against a hostile administrator, prove correctness of supplied metadata or provide an operating-system security sandbox. A signed provider provenance chain is outside the study. Temporal checks are only as informative as the retained source metadata.

2.2 Durable execution and verification

Output directories are reserved before dispatch. Journals preserve attempts, reservations, terminal results and accounting; receipt hashes bind finalized outputs. Measured usage is distinguished from conservative reservations. Recovery inspects retained attempts rather than automatically repeating a potentially billed call. Byte verification and source/semantic replay have separate dispositions; neither is labeled scientific validation. Archived version-one runs retain their original, more limited verification guarantees.

For the paper experiment, a separately frozen runner records all scheduled outputs, guard results, decisions and resource checkpoints. It forbids automatic retries and retains failures. A separate auditor reconstructs native facts from original source rows, rechecks the controls and reconciles the complete schedule. This audit uses a distinct projection implementation rather than treating the producer's expected values as its oracle. Authoring and audit agents are software collaborators, not external expert adjudicators or statistically independent judges.

2.3 Typed reporting and scoring

A report fact contains a stable fact identifier, typed value, unit, source binding and unknown reason. Bindings identify the full source file hash, release, physical row locator and field. Task identity includes hypothesis scope. Booleans cannot impersonate numerical measurements. Native null, a field absent from schema and a native empty value have different reasons. Incomplete collection cannot support a biological absence statement.

The source compiler reconstructs expected facts from originals. Generation receives source-only rows and value-free fact bindings, without expected-reference values. References and evaluation artifacts remain evaluator-only. In-process verified objects are immutable; serialized results acquire their guarantees through replay, not by trusting their own stored status. This is procedural and programmatic isolation, not proof of independent model errors or a hostile-process barrier.

3. Study design

3.1 Source frame and selection

We pinned Open Targets 25.09 evidence, credible-set, target and disease files [10] and the locally downloaded HGNC normalization snapshot dated 10 October 2026 [11]. The HGNC date is a retained acquisition designation, not a provider-certified annotation release. The frozen frame contains 137 original files totaling 2,866,248,239 bytes: 100 GWAS evidence files, 25 credible-set files, ten target files, one disease file and one HGNC file. All source bytes and download receipts were verified. The complete credible-set input contains 2,862,043 physical rows.

The registry enumerates 1,753,850 physical GWAS evidence rows. Its fixed identity, temporal, mapping and eligibility rules retain 1,681,738 rows and exclude 72,112; individual exclusion reasons can overlap. A deterministic identity-only rank over target, disease and locus selects twelve units. Nine previously used target symbols are excluded. Selection does not rank numerical values, source answerability or candidate performance. Each retained unit has one exact locus-identifier match in the complete credible-set table. A locus match is a technical join and does not establish shared study, cohort, allele orientation or biological mechanism. Cross-source variant-identifier agreement is also unresolved: this is a locus-only match, not a qualified composite study-and-variant match.

Broader prior project exposure is unknown for all twelve selected units. They are not certified fresh cases or twelve independent studies. One repeated disease-identifier group is recorded; source-study and cohort dependence remain unresolved. After execution their identities and results are exposed development material. We preserve the pre-run exposure register rather than rewriting it with hindsight.

The rank is ascending SHA-256 of the UTF-8 string seed|release|targetId|diseaseId|studyLocusId, with seed preprint-p05-deterministic-v1 and release 25.09. Lexical target, disease, locus and release fields break hash ties. Missing, conflicting or blocked selected units retain their position and disposition; they are not replaced or reranked to improve answerability or results.

3.2 Inputs and independent references

Each unit requests fifteen fields across its original GWAS and matched credible-set rows: eleven identity/linkage fields and four measurement fields. The measurement components are beta, standard error, p-value mantissa and p-value exponent, all in one declared measurement group per unit. We report these native components without calculating a p-value, assigning effect units, interpreting direction or converting provider scores into support weights. Sample size and other context can be retained without becoming graded facts.

Source-only input packets were sealed before reference construction. A separate source-only branch rebuilt the references from eligible originals, without generation outputs, human responses or prior answer-aware derivatives. An independent audit then directly checked all 180 requested facts against 24 distinct original rows. The reference producer and its own replay share algorithms; the additional direct-projection audit is the separate implementation check. These references establish native field fidelity, not adjudicated biomedical truth.

3.3 Arms, policies and constructed scenarios

The two native arms are a deterministic literal assembler and an always-abstain control. The assembler reads only the generation packet and copies each requested native field with its binding and missingness reason. The abstention arm emits no assertions. No model is run on these twelve units. Native literal outputs receive three policy projections; native abstentions receive the full policy only. Fourteen constructed generic-JSON scenarios receive all three policies. The resulting schedule contains 38 distinct outputs and 90 policy decisions.

The full policy requires five guards: G0 retained-byte/hash anchoring; G1 source-role and ancestor admissibility; G2 the P05 mapping, temporal and inventory wrapper; G3 the base source compiler; and G4 complete exact reporting with output-schema checks. WITHOUT_WRAPPER omits only G2, preserving the base guards. WIRE_ONLY requires G0 and the G4 schema check. These policies are projections over retained guard outcomes, not independent full executions or recommended scientific methods. Source admission through G0-G3 and report exactness at G4 are separate outcomes.

The constructed set has four technically valid and ten deliberately invalid scenarios. Faults cover changed source bytes, evaluator-only ancestry, future release metadata, incomplete inventory, altered number, wrong locator, wrong unknown reason, duplicate JSON keys, Boolean measurement and oversized output. The valid controls include explicit missingness, multiple groups and an entirely unavailable measurement group. C02 tests an absent field through the generic JSON/core path; it does not qualify native absent-column handling, because the unchanged P05 Parquet scanner requires the declared measurement columns. One valid control copies a finite negative standard error: its technical fidelity is correct while its scientific interpretation is unqualified. Controls were exposed and qualified before the study. Their results measure behavior on designed examples, not sensitivity, specificity or natural error prevalence.

3.4 Endpoints, accounting and resources

For each answerable record/family/group, coverage is the fraction of exact matched non-null components. Family coverage averages these fractions equally across answerable groups. Native nulls are excluded from reporting denominators and separately counted. An all-unknown group has undefined coverage, not zero or automatic success. Completion requires an eligible answered/partial report, correct collection status, no mechanically identified critical errors and at least 80% primary measurement coverage. Frozen sensitivities use 60% and 100%. Exact-report verification requires status answered and exactly one correct emission of every declared fact, including correct unknown reasons. A partial report may satisfy coverage but cannot be exact under this stricter rule.

The primary accounting includes every scheduled output and decision. Failed, malformed, interrupted, timed-out or unexecuted cells cannot disappear from a favorable subset. A core-scored subset is secondary and explicitly labeled. We use descriptive counts and paired decision tables only. We report no confidence interval, power calculation, significance test or independent-study inference for this selected set.

The execution protocol was frozen in a committed READY file before native output construction. It binds runtime and extra code pins, sources, receipts, task and reference seals, denominator register, fictional qualification, schedule, budgets and analyzer. Limits include 3,600 seconds for execution and a separate 3,600

seconds for audit, 2 GiB checkpoint process memory and 256 MiB study output. Logical completed hash reads, row enumerations, CPU, wall time and memory checkpoints are retained. Repeated hash reads are logical work, not measured physical disk traffic. No new model call, network transfer or provider charge occurs in this experiment.

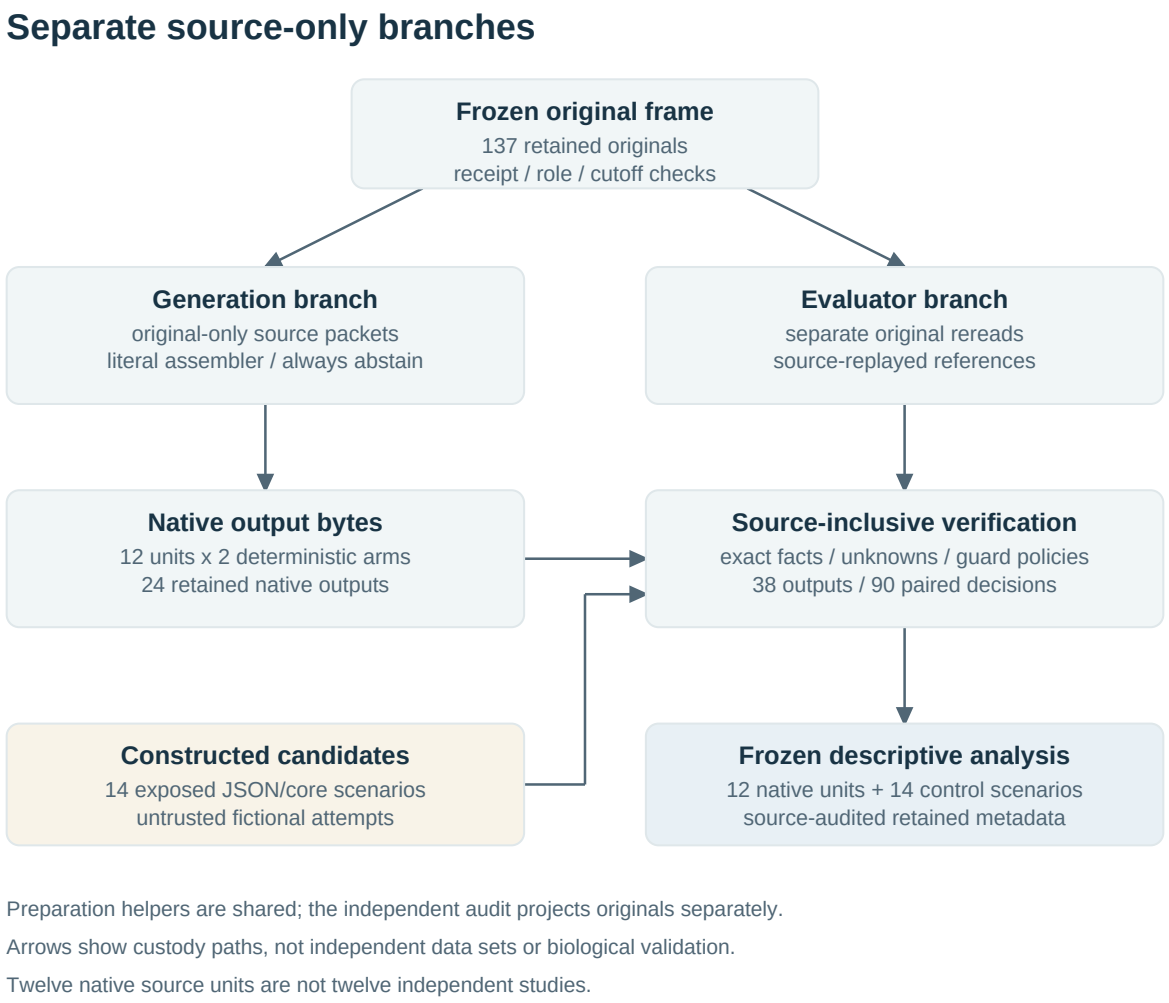


Figure 1. Source-to-report flow and the separate evaluation branch. Shared generation and guard projections are not independent executions.

4. Results

4.1 Native availability and reporting

All twelve native units finalize. Of 180 requested facts, 148 are populated and 32 are native nulls: 108/132 identity slots and 40/48 measurement slots are populated. There are no missing-schema or native-empty measurement slots in this set. One unit has two answerable measurement components, six have three, and five have four. Beta is null in one unit and standard error in seven. Sample size is missing in ten units but is not a graded measurement component.

The literal assembler emits all fifteen requested facts per unit with exact bindings and correct unknown reasons. It achieves verified exact reporting and measurement completion in 12/12 units at each frozen threshold. The abstention control completes 0/12, emitting no fact assertions and incurring no factual mismatch

errors. Abstention therefore provides error avoidance without useful completion. These contrasts establish the expected copy ceiling and the explicit coverage cost of abstention on this task.

Native outcome	Literal assembler	Always abstain
Scheduled outputs	12	12
Verified exact reports	12/12	0/12
Measurement completion at 80%	12/12	0/12
Completion at 60% and 100%	12/12 at both	0/12 at both
Mechanically identified factual errors	0	0
Emitted fact entries	180, including 32 unknowns	0

Source availability of 40/48 measurement slots is distinct from 100% reporting coverage over the answerable components. Missing fields remain unavailable rather than being supplied from another source or inferred from related measurements.



Figure 2. Native source availability, separate from reporting coverage. Forty of forty-eight measurement slots are populated.

4.2 Constructed policy contrasts and full accounting

The full policy admits 0/10 invalid scenarios. WITHOUT_WRAPPER admits 2/10: future-release metadata and incomplete inventory. WIRE_ONLY admits 7/10, demonstrating that a well-formed retained report can remain unsupported under stronger requirements. All three policies reject 0/4 technically valid controls. The all-unknown measurement control remains an exact report with undefined measurement coverage and does not count as completed. All 42 constructed policy decisions are retained.

Constructed outcome	FULL	WITHOUT_WRAPPER	WIRE_ONLY
Invalid scenarios admitted	0/10	2/10	7/10
Technically valid scenarios rejected	0/4	0/4	0/4

The 38-output schedule has 36 returned outputs and two deliberately malformed outputs: duplicate JSON keys and an oversized report. All 90 decisions reach a terminal disposition, with 57 admissions and 33 rejections. The 35 genuinely core-scored outputs are a secondary subset; they are not substituted for the complete schedule. Pairwise constructed-policy discordance is 2/14 for FULL versus WITHOUT_WRAPPER, 7/14 for FULL versus WIRE_ONLY and 5/14 for WITHOUT_WRAPPER versus WIRE_ONLY. These paired differences concern fixed designed controls, not independent clinical observations.

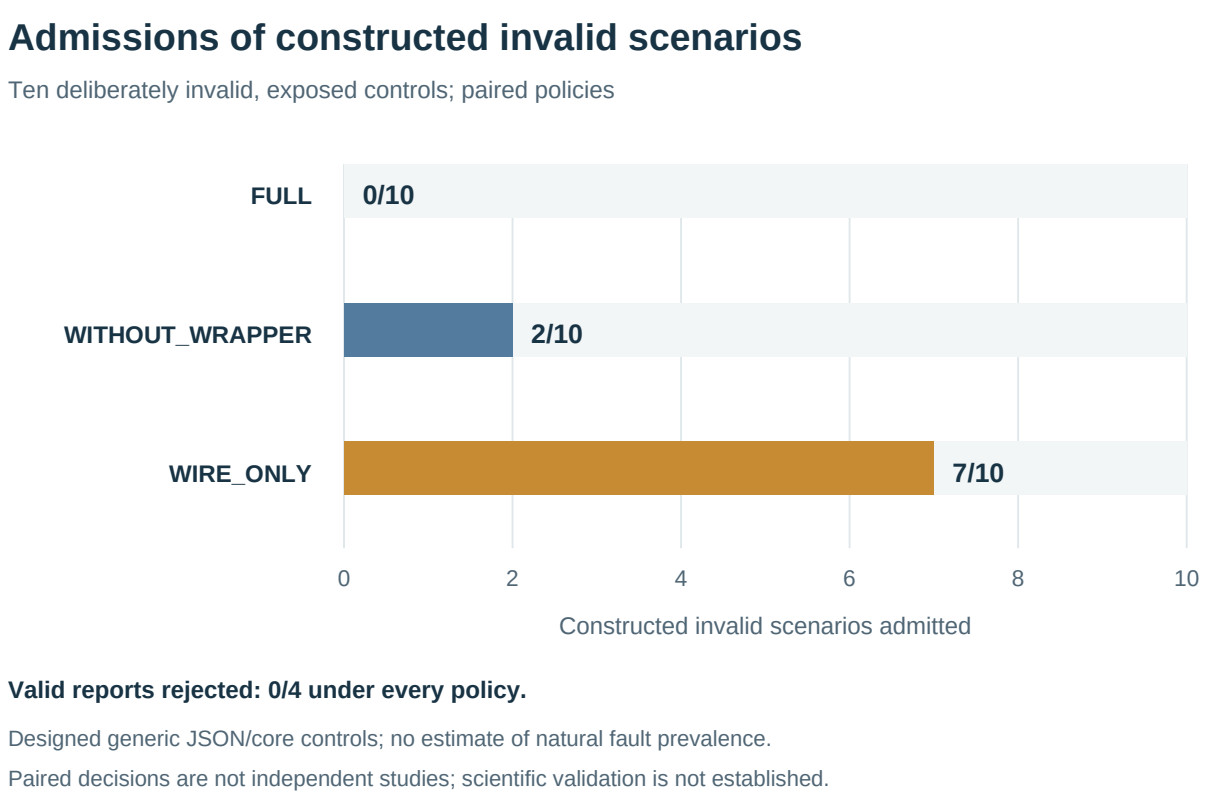


Figure 3. Invalid admissions under the three policies. All fourteen scenarios were constructed and exposed before the study.

4.3 Verification and resource use

The separate source-inclusive auditor passes 2,414 checks with no unresolved finding. It independently reconstructs original-row projections, constructed controls, policy decisions, source proofs and all-output accounting. The frozen analyzer runs once against the pinned ledger and audit. A separate presentation audit passes 351 checks over quantities, tables, captions, data and figure geometry. A fresh-directory local rebuild reproduces all 39 result-bundle files byte-for-byte. Check counts describe implementation verification effort; they are not biomedical sample sizes or an estimate of general correctness.

Production takes 203.828 seconds wall time and 198.984 seconds CPU, with checkpoint-observed process-lifetime peak memory 305,827,840 bytes. The 197.812-second source-verification wrapper is included within production, not an additional phase to sum. The separate audit takes 7.125 seconds and peaks at 333,893,632 bytes. Completed logical hash reads total 140,452,303,784 bytes in production and 2,897,268,566 in the audit; the two phases enumerate source rows twice in total. Physical I/O, partial failed-read bytes and platform costs are unmeasured. Provider spending is $0, which is not a claim of zero overall cost.

Reference construction is a separate preparatory phase: 342.750 seconds, with producer memory unmeasured. Its later audit memory does not recover that missing measurement. Native literal/abstain generation shares a source-packet read per unit, and policy decisions reuse guard outcomes. We do not attribute total process memory to an individual arm or treat projections as independent runtime trials.

4.4 Negative model development, separate from the native study

Before narrowing scope, a frozen twelve-cell development comparison used two literature contexts with deterministic, plain local-model and proposal/check arms plus a controlled missing-input condition. Five actual local-model calls yielded four malformed responses and one output-limit failure; none satisfied the strict report contract. Neither model arm completed either clean context, while the literal baseline completed 2/2. The model was Qwen2.5-14B-Instruct in pinned Q4_K_M GGUF form, served through an ordinary completion interface. Grammar-constrained generation was not used in this probe.

A subsequent predeclared two-call fictional conformance probe passed its two-fact control and failed its eight-fact control on the value/unknown-reason rule. The frozen stop rule closed this model branch. No further prompt retries, model scaling or model evaluation on the twelve native units followed. These are exposed development observations under one model/interface/configuration. They do not identify whether task design, prompting, serialization, model capacity or their interaction caused failure, and they do not establish broad LLM inability. Earlier zero-call setup failures are also retained.

5. Discussion

The study supports a limited engineering conclusion: selected native fields can be reported exactly while eligibility, source ancestry, missingness, failure and resource records remain replayable. Explicit policies make it possible to show which designed faults a weaker policy admits. Exact copying alone already reaches the task's ceiling, so adding an agent cannot demonstrate superior biomedical interpretation on this endpoint. The useful object is the inspectable reporting and verification contract, including its limits.

Prior work supplies substantial adjacent capabilities. The pinned Virtual Biotech implementation supports headless execution and explicitly distinguishes artifact integrity from evidence coverage [12]. Alvessa already verifies retrieved-record statements [4]. TxBench-PP addresses structured preclinical decisions [8], and BiomniBench reports process rubrics [9]; we do not inherit their task validity or judge calibration. Retrieve to Explain evaluates evidence-driven target prediction against different endpoints [13]. None was executed here.

Missing feature assessments are not evidence of competitor deficiencies. Our bounded contribution is therefore an implementation and source-reporting feasibility result, with incremental utility and scientific value still to establish.

The missing context is substantive. Exact locus matches and literal beta/standard-error components do not establish compatible study identity, ancestry, genome build, allele orientation, units, intervention direction or causal effect. A reference assembled from a public database can faithfully reproduce an upstream error. The constructed finite negative standard-error example makes this distinction explicit. A clinical predictor, Mendelian-randomization analysis or therapeutic ranking would require additional data qualification and scientific validation before an outcome study.

6. Limitations

The twelve units were identity-ranked from a local archived subset, not randomly sampled from all biomedical evidence. Their prior exposure is unresolved, source-study independence is unqualified and the selected rows contain sufficient literal measurements. There is no certified holdout set, no generalization interval and no clinical outcome evaluation. Public-download timestamps do not establish historical availability. The missing Open Targets study input prevents upstream study-context verification.

The fourteen controls are known, designed and previously qualified. Passing them establishes only the exercised behavior; unmodeled corruptions, adversarial combinations and real-world error prevalence remain unmeasured. WIRE_ONLY and WITHOUT_WRAPPER are diagnostic projections, not complete alternative agents. Multiple decisions from one output and multiple components from one unit are dependent. The copying baseline is deliberately simple and its perfect performance is expected.

Role separation, code pins and independent projection checks reduce identifiable implementation errors but do not provide an operating-system sandbox, cryptographic provider identity or independent scientific adjudication. AI agents assisted implementation, auditing, literature checking and writing. They are not external experts, and their agreement does not certify biomedical correctness. Historical bespoke responses and answer-aware descendants are excluded from this study rather than reused as references or labels.

The local model results are small negative development probes and cannot support model-family comparisons or a general workflow ranking. The source study uses no model. Full Virtual Biotech reproduction, clinical scoring, historical cohorts, MR, tasks T2-T5 and the H6 reanalysis remain separate uncompleted programmes. Wider project data gaps are not resolved by finishing this technical experiment.

The P06 study and its results and costs remain frozen: primary production took 203.828 seconds and its source audit took 7.125 seconds. A separately versioned, post-result P09 adapter has qualified installed, relocated reconstruction on Windows x64 with CPython 3.11.15 and PyArrow 21.0.0. The native replay used all 137 exact originals; the complete adapter reproduced 38 outputs and 90 decisions across twelve native units and fourteen constructed controls. Reconstruction took 289.938 seconds, and the separate source-inclusive independent audit passed 6,862 checks in 15.187 seconds. The minimal public catalog creates new reference fingerprints; historical equivalence removes only `method.catalog_sha256`. New G0 checks public protocol, code, input and output byte bindings under independently retained authority, without establishing publisher authenticity or verifying old private journals or sealed trees. Fresh exact-byte reacquisition of the 136 external Open Targets originals remains untested. This local technical qualification establishes neither scientific validity nor public availability; author facts remain deferred.

7. Availability and declarations

The repository contains implementation, frozen protocols, retained failures, auditable results, an editable manuscript, bibliography and supplementary methods. At draft time, no public artifact deposit, permanent identifier or publication is claimed. A public release must use an explicit allowlist; original datasets, model weights, credentials, historical response archives and private or outcome-aware artifacts are not implicitly licensed by the repository's Apache-2.0 code license. The supplement identifies local reconstruction requirements and exact study anchors. Local file locations are implementation details, not public access guarantees.

AI assistance: Codex-hosted agents supported software development, source checking, technical auditing and manuscript preparation. The retained scientific report has no external-expert validation. Separate local-model probes and their failures are disclosed above. Platform-model identity, hidden compute and platform cost are not fully metered. No AI system is an author; human authors must verify and accept responsibility for the final text and disclosures.

Author names, affiliations, correspondence, contributions, funding, competing interests, acknowledgments and any required ethics/data-governance declarations require owner-confirmed facts. They are unresolved submission metadata, not requests for external scientific experts. No statement of absent funding, conflicts or oversight is inferred from missing information. No patients were recruited or clinical recommendations issued by this technical experiment.

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Supplementary methods and artifact guide

Companion to the working AsOf source-reporting draft, 1 November 2026. Public deposit and owner declarations remain pending. The separately qualified local artifact is described in S8. Paths below describe this local research checkout, not a public data deposit. No external expert response is required or used.

S1. Frozen chronology and immutable anchors

All dates below are UTC. The information cutoff is 2026-11-01T07:16:26.101448+00:00. The run used source-only inputs sealed before references; references were finalized before generation. The final experiment READY was committed as 31c52d3 before the single native study. Later audits and analysis do not overwrite its ledger or pre-run exposure register.

Stage	UTC output directory suffix	Status
Prospective registry	20261101T071713Z	Twelve identity-ranked units; complete physical-row enumeration
Original-only packets	20261101T074942Z	Twelve source-only packets; zero blocked
Neutral tasks	20261101T075541Z	Twelve tasks; no evaluator fact values
References	20261101T081933Z	Twelve evaluator-only references; independently source checked
Exposure/denominator register	20261101T083548Z	Pre-run UNKNOWN broader exposure; frozen answerability
Fictional source controls	20261101T090035Z	Fourteen source-only fixture scenarios
Fictional qualification	20261101T090042Z	Fourteen scenarios/42 decisions; no native outputs
Actual study	20261101T091345Z	Thirty-eight outputs/90 decisions, terminal
Frozen descriptive analysis	20261101T092110Z	Pinned ledger and later source-audit gate

Exact file SHA-256 anchors are listed below. These are byte hashes, not canonical-object fingerprints. The distinction matters because whitespace changes affect byte hashes. The manifest and audit objects record additional canonical fingerprints where used.

Artifact	Exact SHA-256
EXPERIMENT-READY-v1.json	49c2ce672356bfcfd572a7b6c963c56bd5c96e3d042a9972f2c36434eb b8de3f
Study ledger.json	b2a9727ca9ce89865b80ffe80b21a325ea6d91d8844247740ac2c2ecc 9a4faa0
Study seal.json	84b1e87b187dd5704c143f0da0fcac41103a88affe8c8b022c2890c8dc 89c70f
EXPERIMENT-AUDIT-v1.json	f10f15959f8497821115f65268808c98ae345079c7361722390a479b15 506922

Artifact	Exact SHA-256
Analysis analysis.json	ee9f02ea3c3692bbd8b7d63d2d2cd545cfd769ccee9ee7454b4ed00c7ac87312
Results ARTIFACTS.json	84d4a31e842d3085133d384429729ef2501425d33bf2b92f65d9c64699a87428
ANALYSIS-PRESENTATION-AUDIT-v1.json	80815c8fafd7e2225930f3ea54fbccdca926269c872f893a9f841386c14ce031

READY, source-audit and presentation-audit files live under planning/review-assets/preprint-p05-20261031/. The study and analysis directories live under data/derived/ with their UTC suffixes. The original ledger's COMPLETED_PENDING_INDEPENDENT_AUDIT status remains unchanged; the separately pinned later PASS audit supplies the subsequent verification result. A successful audit is not a reason to rewrite old output bytes.

S2. Source and interpretation card

Product	Local selected frame	Purpose	Interpretation limit
Open Targets 25.09 GWAS evidence	100 files; 1,753,850 rows	Target/disease/locus identity and native fields	Provider aggregation and publication dates do not establish causal relevance
Open Targets 25.09 credible sets	25 files; 2,862,043 rows	Exact locus join and four native measurement components	No inferred effect units, allele orientation, ancestry or causal direction
Open Targets 25.09 targets	Ten files; 78,726 rows	Canonical target normalization	Current normalization cannot silently fill historical gaps
Open Targets 25.09 diseases	One file; 39,530 rows	Required disease identities	Ontology identity does not establish disease-model equivalence
HGNC local 10 October 2026 snapshot	One TSV; 45,233 rows	Approved symbol/Ensembl normalization	Acquisition date differs from annotation release and historical availability

All 137 originals total 2,866,248,239 bytes. Receipt copies are content-addressed and archived separately from transport receipts. Parent paths, source hashes, row-group/row locators and role restrictions are retained. Full original-file hashes cannot be reproduced by an extract containing only the 24 used rows. Source inventories are selected subsets of the providers' overall releases.

Source-valued InputGraph parents are limited to pinned eligible originals, with their retained receipt proofs. Validated neutral Task/Snapshot, selected-case routing and plan configuration remain outside the source-valued InputGraph; their hashes are checked as configuration, and they cannot be relabeled as original evidence. Registry captures, mapping/coverage audits, SQL values, earlier generation packets, expected references, model outputs, old selected-join descendants and historical answer-aware derivatives are quarantined from the new source-only reference branch. Generation uses its separately admitted original-only packets and value-free fact bindings; evaluator references and descendants cannot become generation evidence. Recorded parent roles, exact byte anchors and fresh original projections enforce the declared branch policy; they do not prove a hostile operating-system sandbox.

The complete Open Targets study table is not available in the eligible local input set. Twelve exact locus matches therefore do not qualify upstream study agreement. GWAS-versus-credible-set variant-identifier agreement is unresolved; the join is not a qualified composite study-and-variant match. Genome build, ancestry, units, effect alleles and cohort overlap remain unresolved. Sample size is unavailable in ten selected credible-set records and is not a graded measurement fact. Numerical clinical features are empty and probabilities null. No MR estimate, colocalization probability, intervention direction or effect-size support weight is calculated.

Source availability is separate from scientific validity. Identity 108/132 and measurement 40/48 are populated-field counts. Nulls 24+8 are preserved. A populated source value can be scientifically problematic; a null cannot be replaced with zero. No claim that missing local evidence establishes biological absence is allowed.

S3. Selection, exposure and dependence

The registry scans all 1,753,850 local GWAS physical rows. Fixed rules retain 1,681,738 eligible rows and exclude 72,112. Eligibility reasons can overlap, so a summed reason histogram is not the number of excluded rows. Target/disease/locus identity rank is fixed without measurement-value or candidate-performance ranking. The nine excluded previously used symbols are CD276, OSMR, ERBB2, PCSK9, IL6R, CFTR, GLP1R, SCN9A and CTLA4.

For each eligible unique release/target/disease/locus unit, the rank key is SHA-256 of the UTF-8 concatenation preprint-p05-deterministic-v1|25.09|targetId|diseaseId|studyLocusId. Ascending hash order is followed by lexical targetId, diseaseId, studyLocusId and release_id tie order; the first twelve units are selected. Missing, conflicting, blocked or insufficient selected units keep their position, failure or missingness. There is no replacement, answerability reranking or numeric-performance selection after the freeze.

The units are p05-01 through p05-12, with one original GWAS and one exact-locus credible-set row per unit. The evaluator register contains 180 requested slots: 132 identity and 48 measurement. Native nulls are 24 identity and eight measurement. Per-unit answerable measurement group sizes are 3,3,4,4,4,3,3,2,4,3,3,4. Identity and measurement components are not independent observations. Twelve locus strings and one repeated disease string group do not establish twelve independent studies or cohorts.

Broader exposure is UNKNOWN in the frozen pre-run register. After the study all twelve outputs and underlying identities are exposed. New holdout claims cannot be created by moving them to a different folder. Old bespoke review responses, normalized responses, candidate assessments, answer-aware mappings and descendants are archive-only and excluded from generation, reference construction, judging, tuning and source/case/arm selection. This exclusion is a declared input policy with retained ancestry checks, not evidence that an AI model never encountered a public source during training.

S4. Fields, wire contract and endpoint

Each output declares schema_version, task_id, task_sha256, status, facts, collection_status and reason. A fact declares fact_id, value, unit, source and unknown_reason. The source object binds source_id, file sha256, release_id, physical locator and field. No prompt-only prose is scored as an emitted typed fact. Arrays preserve order, integers/floats/Booleans remain distinct, duplicate JSON keys are rejected and oversized outputs fail without truncation or repair.

Missingness reasons distinguish field_not_in_schema, null_source_value and empty_source_value. Native empty values can be reportable or unknown only according to the declared field policy. Unit is null for these native reporting tasks. An abstained or insufficient_input response has no facts and a reason. Wrong collection

status or a wrong value, unit, binding or missingness reason is a critical mechanical error. Unknown fact IDs and duplicate fact emissions are errors.

C02's absent-standard-error example tests the generic JSON/core path only. The unchanged P05 native Parquet scanner requires its declared measurement columns, so that control does not qualify native absent-column acceptance or weaken the native schema policy. All eight unavailable native measurement slots in this study are present-column nulls.

For an answerable group g, let n_g be its number of populated reference components and m_g the number of exactly matched emitted components. Group coverage is m_g/n_g. Family coverage is the equal-weight mean over groups with n_g greater than zero. Measurement completion requires an eligible answered/partial response, correct collection status, no critical mechanical errors and mean measurement coverage at least 4/5. Sensitivities use 3/5 and 1. A zero-answerable-group family has undefined coverage and cannot pass by 0/0. The literal twelve-unit study has one answerable measurement group per unit. Technically valid C04 has none and remains uncompleted despite exact reporting.

The four native measurement fields are beta, standardError, pValueMantissa and pValueExponent. P-value mantissa/exponent are reported separately; no p-value is recomputed. Exactness uses typed source-value equality, not a scientific tolerance or inferred sign. Field-family coverage and exact all-fact reporting are distinct. The missing-schema/empty behavior occurs in constructed controls and regressions; it is not falsely reported as observed native missingness.

Strict exactness additionally requires a returned, structurally scorable report with status answered, zero critical errors and exactly one emission of each declared slot. Status partial is permitted by the coverage-completion contract but cannot satisfy strict exactness even with full coverage. Reference-value/null correctness, source bindings, units, unknown reasons and collection status must all agree.

S5. Fixed guards and control inventory

ID	Constructed condition	First full-policy failure
C01	Complete valid report	None
C02	Valid null/absent/empty handling	None
C03	Multiple groups and finite negative standard error, interpretation unqualified	None
C04	All measurements unavailable, exact report, coverage undefined	None
F01	Changed source hash	G0
F02	Evaluator-only ancestor	G1
F03	Future release metadata	G2
F04	Incomplete declared inventory	G2
F05	Wrong number	G4
F06	Wrong physical locator	G4
F07	Wrong unknown reason	G4
F08	Duplicate JSON key	G4/schema
F09	Boolean in a measurement field	G3
F10	Oversized output	G4/schema

FULL requires G0-G4. WITHOUT_WRAPPER requires G0,G1,G3,G4. WIRE_ONLY requires G0 and G4's schema status, not G4's exactness status. An unexecuted guard is not automatically a pass. F02 and F09 may still be admitted by WIRE_ONLY because their required schema/byte checks pass while stronger source checks do not. No policy is credited with factual source fidelity solely because the wire is valid. All intended decisions remain in the ledger, including early guard failures.

There are 24 native outputs: twelve literal plus twelve abstentions. Literal outputs have 36 decisions and abstentions twelve. Fourteen controls have 42 decisions. Total: 38 outputs and 90 decisions. All 90 reach a terminal verdict; 57 admit and 33 reject under their respective policies. Returned36 and deliberately malformed2 is output accounting, not decision accounting. The 35 core-scored-output subset is secondary. These denominators must not be mixed.

S6. Development model card and stopped routes

The local development model is Qwen/Qwen2.5-14B-Instruct-GGUF, Q4_K_M, revision b466e1f8c07172155743e8e1307507d8a4f91fbd, served by the pinned b11717 runtime. Three weight-shard hashes are retained in the P04 protocol. No weights are redistributed by this manuscript. The configured context is 16,384 tokens, eight threads, seed zero and at most 2,048 output tokens per call. The configured GPU-layer count is 99; actual hardware/offload is not inferred from this setting. Ordinary completion is used without grammar/schema-constrained decoding.

P04's two exposed literature contexts have no populated graded measurement fields. Their model contract failures cannot be described as failed native GWAS extraction. Its twelve scheduled cells yield seven returned, four malformed and one failed cell; five actual model calls consume 13,582 input and 5,726 output tokens. Wall time is 424.516 seconds. Plain and proposal/check arms each complete 0/2 clean contexts; literal completes 2/2. Two preceding zero-call setup failures and later metadata corrections remain retained.

P04b's two fictional calls consume 2,501 input and 1,881 output tokens over 38.563 seconds. The two-fact example passes and the eight-fact example is malformed. This fails the fixed branch-qualification rule and closes model scaling. The observed failures are conditional on the task, prompt, format, implementation and model. They are not a comparison of model families, a clinical benchmark or an explanation of failure causality. No new generation is authorized by preparing the paper.

S7. Verification, failures and resource interpretation

Preparation audits retain 789 registry checks, 1,238 input checks, 1,163 reference checks and a separate 478-check root register audit. These are bounded software/source consistency checks with overlapping evidence, not additive independent validation samples. Final runner, auditor and analysis preparation suites contain 36, 25 and 17 methods respectively. A keyed-dictionary/list mismatch in the independent auditor was reproduced in a retained regression and fixed before the actual study. No actual native study was run under that defective auditor.

Production has one original-row enumeration, 203.828 seconds wall and 198.984375 seconds CPU. Sampled RSS is 241,364,992 bytes and checkpoint-observed lifetime process peak is 305,827,840 bytes. Source verification is an overlapping 197.812-second subphase. Separate audit has one enumeration, 7.125 seconds wall, 7.046875 seconds CPU, sampled RSS 327,589,888 bytes and peak 333,893,632 bytes. Combined completed logical hash reads are 143,349,572,350 bytes. No physical I/O or dollar-valued platform cost is inferred from this.

The reference producer took 342.750 seconds and did not measure RSS/peak because its optional instrumentation was absent. This null is retained. Later process measurements cannot substitute for it. Source-preparation costs, study costs, audit costs and post-result rendering costs are separate. Shared native packet reads and policy projections are counted once at their actual operation granularity.

The 351-check presentation audit initially flagged four checker differences concerning unavailable scores, a scope label, resource-label spacing and floating SVG serialization. The checker was corrected; paper result payloads were unchanged. The initial report/code remain retained. A fresh local rebuild matches all 39 result-bundle files under the same runtime/standard-font environment. This does not guarantee cross-platform byte identity or full-study relocation.

S8. Result files and reproduction scopes

The paper/results bundle includes eleven CSV tables totaling 392 rows, eleven readable Markdown views with 316 displayed rows, three vector PDF/SVG figures, PNG previews, exact figure data and captions, plus ARTIFACTS.json. The CSVs retain full serialized numeric precision; Markdown rounds display values. Unknown cells stay unknown rather than zero. The manifest binds every generated payload to its exact analyzed source. See paper/results/README.md for the local rebuild command and resource semantics.

The frozen descriptive analyzer is tools/science/analyze_source_validation_experiment.py. It consumes the exact audited ledger and produces the sealed analysis; it does not reread protected outcomes or use new judgments. tools/science/build_p05_paper_assets.py formats the already frozen quantities. Rendering was written after results and is not represented as preregistered analysis. Bibliographic source verification is in

paper/RELATED-WORK-SOURCE-AUDIT.md, with machine-readable paper/references.bib and paper/references.json.

Verified scopes now also include the separately versioned P09 installed, relocated reconstruction and its source-inclusive independent audit on Windows x64, CPython 3.11.15 and PyArrow 21.0.0. These are post-result technical qualification operations, separate from the frozen P06 study and its 203.828-second production / 7.125-second source audit. Reconstruction reproduced all 38 output hashes and 90 decisions in 289.938 seconds; the separate audit passed 6,862 checks in 15.187 seconds. All 137 exact originals remain mandatory: 136 Open Targets files are external, and the one exact CC0 HGNC file is included with its receipt. Selected-row extracts cannot replace them. The public catalog produces genuine new reference identities; equivalence removes only `method.catalog_sha256`, while new G0 does not establish publisher authenticity or old private-archive integrity. Fresh exact-byte reacquisition of the 136 external originals remains untested. Policy-session logical counters exclude repeated compiler/caller reads; total physical I/O and platform cost remain unknown. The frozen P09 runtime bundle retains its earlier paper snapshot; later P10 working-paper and PDF updates are separate versions.

S9. Availability, rights and unresolved author metadata

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